Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
graph a linear equation by plotting points.			
graph vertical and horizontal lines.			

ⓑ *After reviewing this checklist, what will you do to become confident for all objectives?*

11.3 Graphing with Intercepts

Learning Objectives

By the end of this section, you will be able to:

- Identify the intercepts on a graph
- Find the intercepts from an equation of a line
- Graph a line using the intercepts
- Choose the most convenient method to graph a line

☑ **BE PREPARED** 11.7 Before you get started, take this readiness quiz.

Solve: $3x + 4y = -12$ for x when $y = 0$.
If you missed this problem, review Example 9.62.

☑ **BE PREPARED** 11.8 Is the point $(0, -5)$ on the x-axis or y-axis?
If you missed this problem, review Example 11.5.

☑ **BE PREPARED** 11.9 Which ordered pairs are solutions to the equation $2x - y = 6$?
ⓐ $(6, 0)$ ⓑ $(0, -6)$ ⓒ $(4, -2)$.
If you missed this problem, review Example 11.8.

Identify the Intercepts on a Graph

Every linear equation has a unique line that represents all the solutions of the equation. When graphing a line by plotting points, each person who graphs the line can choose any three points, so two people graphing the line might use different sets of points.

At first glance, their two lines might appear different since they would have different points labeled. But if all the work was done correctly, the lines will be exactly the same line. One way to recognize that they are indeed the same line is to focus on where the line crosses the axes. Each of these points is called an **intercept of the line**.

Intercepts of a Line

Each of the points at which a line crosses the x-axis and the y-axis is called an intercept of the line.

Let's look at the graph of the lines shown in Figure 11.14.

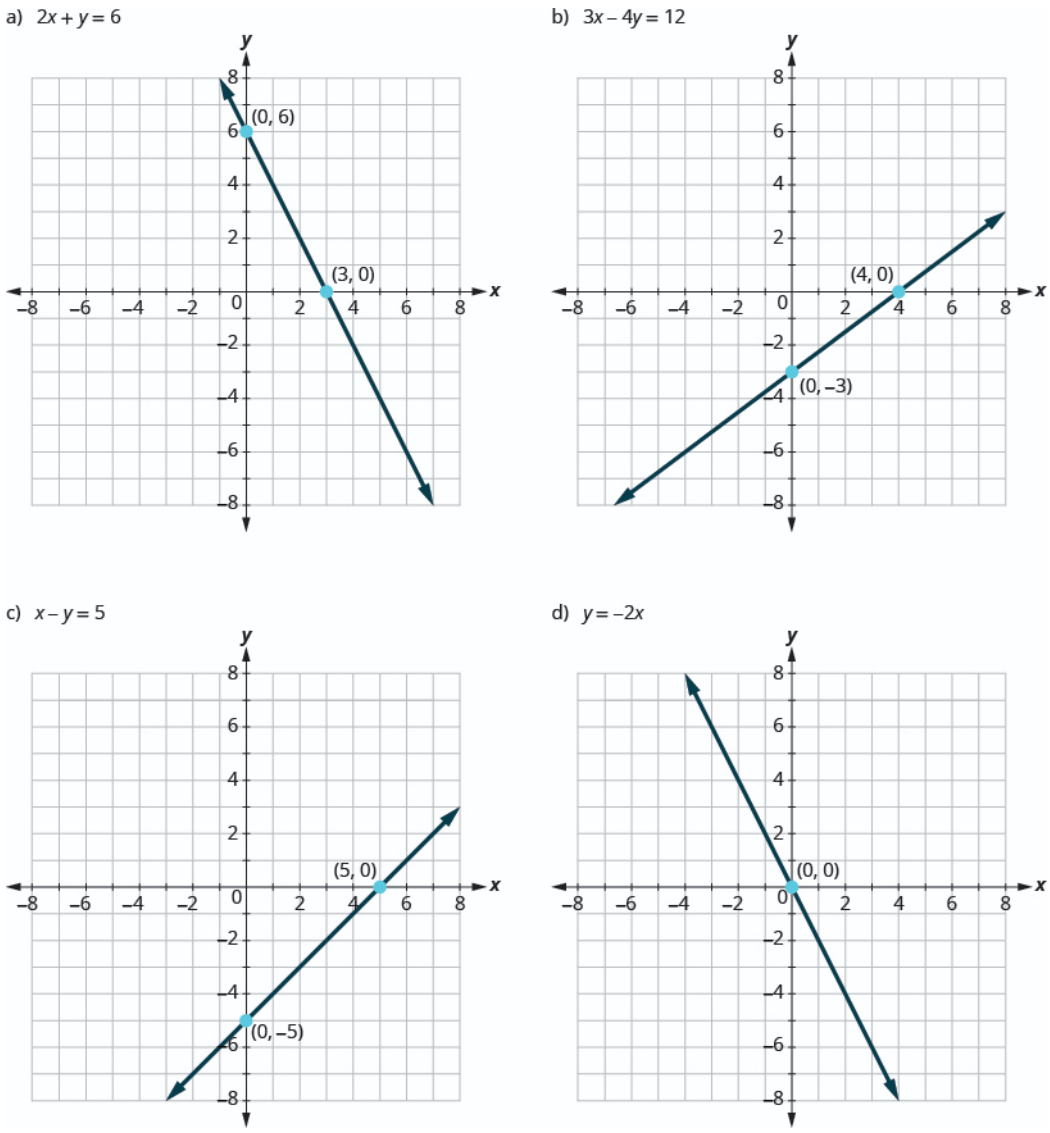


Figure 11.14

First, notice where each of these lines crosses the *x*- axis:

Figure:	The line crosses the x-axis at:	Ordered pair of this point
42	3	(3,0)
43	4	(4,0)
44	5	(5,0)
45	0	(0,0)

Do you see a pattern?

For each row, the *y*- coordinate of the point where the line crosses the *x*- axis is zero. The point where the line crosses the *x*- axis has the form $(a, 0)$; and is called the *x-intercept* of the line. The **x-** intercept occurs when y is zero.

Now, let's look at the points where these lines cross the y-axis.

Figure:	The line crosses the y-axis at:	Ordered pair for this point
42	6	(0,6)
43	-3	(0,-3)
44	-5	(0,-5)
45	0	(0,0)

x- intercept and y- intercept of a line

The x-intercept is the point, $(a, 0)$, where the graph crosses the x-axis. The x-intercept occurs when y is zero.

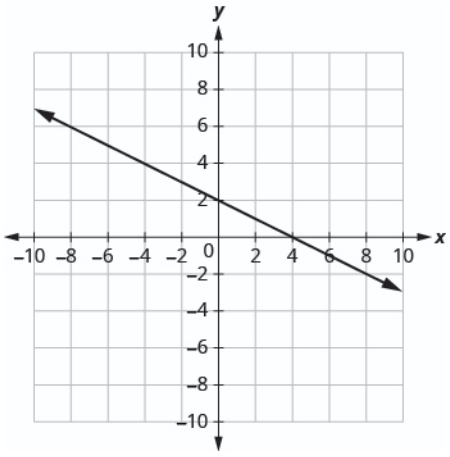
The y-intercept is the point, $(0, b)$, where the graph crosses the y-axis.

The y-intercept occurs when x is zero.

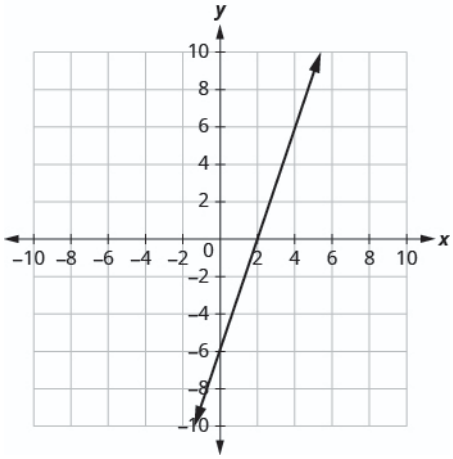
EXAMPLE 11.23

Find the x- and y-intercepts of each line:

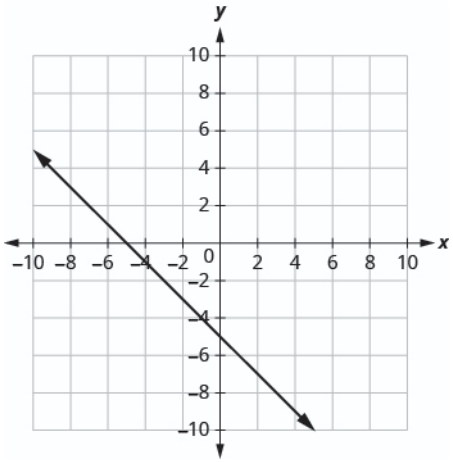
ⓐ $x + 2y = 4$



ⓑ $3x - y = 6$



ⓒ $x + y = -5$



Solution

ⓐ

The graph crosses the x-axis at the point (4, 0).	The x-intercept is (4, 0).
The graph crosses the y-axis at the point (0, 2).	The y-intercept is (0, 2).

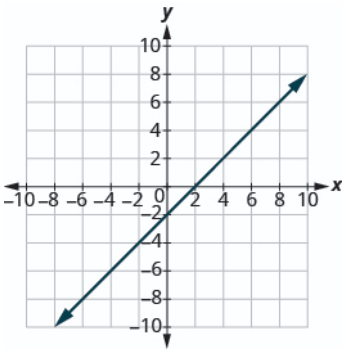
ⓑ

The graph crosses the x-axis at the point (2, 0).	The x-intercept is (2, 0)
The graph crosses the y-axis at the point (0, −6).	The y-intercept is (0, −6).

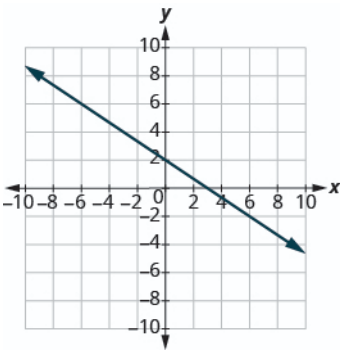
ⓒ

The graph crosses the x-axis at the point (−5, 0).	The x-intercept is (−5, 0).
The graph crosses the y-axis at the point (0, −5).	The y-intercept is (0, −5).

TRY IT 11.44 Find the x- and y-intercepts of the graph: $x - y = 2$.



TRY IT 11.45 Find the x- and y-intercepts of the graph: $2x + 3y = 6$.



Find the Intercepts from an Equation of a Line

Recognizing that the x-intercept occurs when y is zero and that the y-intercept occurs when x is zero gives us a method to find the intercepts of a line from its equation. To find the x-intercept, let $y = 0$ and solve for x. To find the y-intercept,

let $x = 0$ and solve for y.

Find the *x* and *y* from the Equation of a Line

Use the equation to find:

- the *x*-intercept of the line, let $y = 0$ and solve for *x*.
- the *y*-intercept of the line, let $x = 0$ and solve for *y*.

x	y
	0
0	

EXAMPLE 11.24

Find the intercepts of $2x + y = 6$

Solution

We'll fill in Figure 11.15.

$2x + y = 6$		
x	*y*	
	0	*x*-intercept
0		*y*-intercept

Figure 11.15

To find the x- intercept, let $y = 0$:

	$2x + y = 6$
Substitute 0 for *y*.	$2x + 0 = 6$
Add.	$2x = 6$
Divide by 2.	$x = 3$
The *x*-intercept is (3, 0).	

To find the y- intercept, let $x = 0$:

	$2x + y = 6$
Substitute 0 for *x*.	$2 \cdot 0 + y = 6$
Multiply.	$0 + y = 6$

Add.	$y = 6$
The y-intercept is (0, 6).	

$2x + y = 6$	
x	y
3	0
0	6

Figure 11.16

The intercepts are the points $(3, 0)$ and $(0, 6)$.

TRY IT 11.46 Find the intercepts: $3x + y = 12$

TRY IT 11.47 Find the intercepts: $x + 4y = 8$

EXAMPLE 11.25

Find the intercepts of $4x-3y = 12$.

Solution

To find the x-intercept, let $y = 0$.

	$4x - 3y = 12$
Substitute 0 for y.	$4x - 3 \cdot 0 = 12$
Multiply.	$4x - 0 = 12$
Subtract.	$4x = 12$
Divide by 4.	$x = 3$

The x-intercept is $(3, 0)$.

To find the y-intercept, let $x = 0$.

	$4x - 3y = 12$
Substitute 0 for x.	$4 \cdot 0 - 3y = 12$
Multiply.	$0 - 3y = 12$
Simplify.	$-3y = 12$
Divide by –3.	$y = -4$

The y-intercept is $(0, -4)$.

The intercepts are the points $(-3, 0)$ and $(0, -4)$.

$4x-3y=12$	
x	**y**
3	0
0	−4

TRY IT 11.48 Find the intercepts of the line: $3x-4y=12$.

TRY IT 11.49 Find the intercepts of the line: $2x-4y=8$.

Graph a Line Using the Intercepts

To graph a linear equation by plotting points, you can use the intercepts as two of your three points. Find the two intercepts, and then a third point to ensure accuracy, and draw the line. This method is often the quickest way to graph a line.

EXAMPLE 11.26

Graph $-x+2y=6$ using intercepts.

Solution

First, find the x-intercept. Let $y=0$,

$$\begin{aligned} -x+2y&=6\\ -x+2(0)&=6\\ -x&=6\\ x&=-6 \end{aligned}$$

The x-intercept is $(-6,0)$.

Now find the y-intercept. Let $x=0$.

$$\begin{aligned} -x+2y&=6\\ -0+2y&=6 \end{aligned}$$

$$\begin{aligned} 2y&=6\\ y&=3 \end{aligned}$$

The y-intercept is $(0,3)$.

Find a third point. We'll use $x=2$,

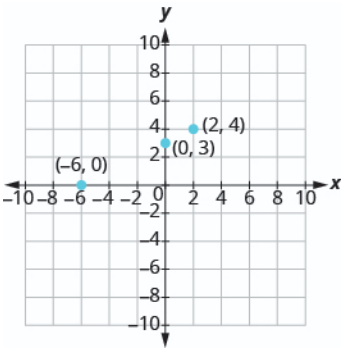
$$\begin{aligned} -x+2y&=6\\ -2+2y&=6 \end{aligned}$$

$$\begin{aligned} 2y&=8\\ y&=4 \end{aligned}$$

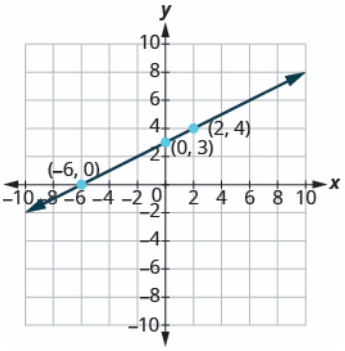
A third solution to the equation is $(2, 4)$.

Summarize the three points in a table and then plot them on a graph.

$-x + 2y = 6$		
x	**y**	**(x,y)**
-6	0	$(-6, 0)$
0	3	$(0, 3)$
2	4	$(2, 4)$



Do the points line up? Yes, so draw line through the points.



TRY IT 11.50 Graph the line using the intercepts: $x-2y = 4$.

TRY IT 11.51 Graph the line using the intercepts: $-x + 3y = 6$.

HOW TO

Graph a line using the intercepts.

Step 1. Find the $x-$ and y-intercepts of the line.
- Let $y = 0$ and solve for x
- Let $x = 0$ and solve for y.

Step 2. Find a third solution to the equation.

Step 3. Plot the three points and then check that they line up.

Step 4. Draw the line.

EXAMPLE 11.27

Graph $4x-3y=12$ using intercepts.

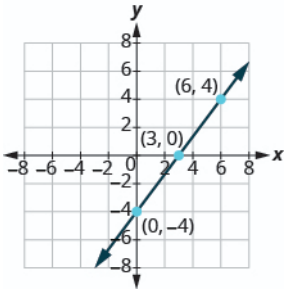
Solution

Find the intercepts and a third point.

x-intercept, let $y=0$	y-intercept, let $x=0$	third point, let $y=4$
$4x-3y=12$	$4x-3y=12$	$4x-3y=12$
$4x-3(0)=12$	$4(0)-3y=12$	$4x-3(4)=12$
$4x=12$	$-3y=12$	$4x-12=12$
$x=3$	$y=-4$	$4x=24$
		$x=6$

We list the points and show the graph.

$4x-3y=12$		
x	y	(x,y)
3	0	$(3,0)$
0	-4	$(0,-4)$
6	4	$(6,4)$



TRY IT 11.52 Graph the line using the intercepts: $5x-2y=10$.

TRY IT 11.53 Graph the line using the intercepts: $3x-4y=12$.

EXAMPLE 11.28

Graph $y=5x$ using the intercepts.

Solution

x-intercept; Let $y = 0$.	y-intercept; Let $x = 0$.
$y = 5x$	$y = 5x$
$0 = 5x$	$y = 5(0)$
$0 = x$	$y = 0$
$x = 0$	The y-intercept is (0, 0).

The x-intercept is (0, 0).

This line has only one intercept! It is the point $(0, 0)$.

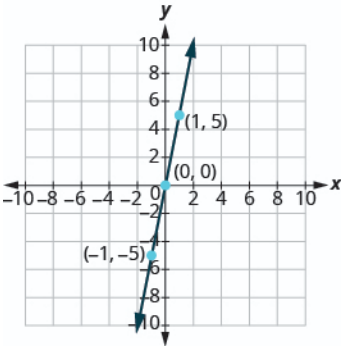
To ensure accuracy, we need to plot three points. Since the intercepts are the same point, we need two more points to graph the line. As always, we can choose any values for x, so we'll let x be 1 and -1.

$x = 1$	$x = -1$
$y = 5x$	$y = 5x$
$y = 5(1)$	$y = 5(-1)$
$y = 5$	$y = -5$
(1, 5)	(–1, –5)

Organize the points in a table.

$y = 5x$		
x	y	(x, y)
0	0	$(0, 0)$
1	5	$(1, 5)$
-1	-5	$(-1, -5)$

Plot the three points, check that they line up, and draw the line.



TRY IT 11.54 Graph using the intercepts: $y = 4x$.

TRY IT 11.55 Graph using the intercepts: $y = -x$.

Choose the Most Convenient Method to Graph a Line

While we could graph any linear equation by plotting points, it may not always be the most convenient method. This table shows six of equations we've graphed in this chapter, and the methods we used to graph them.

	Equation	Method
#1	$y = 2x + 1$	Plotting points
#2	$y = \frac{1}{2}x + 3$	Plotting points
#3	$x = -7$	Vertical line
#4	$y = 4$	Horizontal line
#5	$2x + y = 6$	Intercepts
#6	$4x - 3y = 12$	Intercepts

What is it about the form of equation that can help us choose the most convenient method to graph its line?

Notice that in equations #1 and #2, *y* is isolated on one side of the equation, and its coefficient is 1. We found points by substituting values for *x* on the right side of the equation and then simplifying to get the corresponding *y*- values.

Equations #3 and #4 each have just one variable. Remember, in this kind of equation the value of that one variable is constant; it does not depend on the value of the other variable. Equations of this form have graphs that are vertical or horizontal lines.

In equations #5 and #6, both *x* and *y* are on the same side of the equation. These two equations are of the form $Ax + By = C$. We substituted $y = 0$ and $x = 0$ to find the *x*- and *y*- intercepts, and then found a third point by choosing a value for *x* or *y*.

This leads to the following strategy for choosing the most convenient method to graph a line.

HOW TO

Choose the most convenient method to graph a line.

Step 1. If the equation has only one variable. It is a vertical or horizontal line.
- $x = a$ is a vertical line passing through the x-axis at a
- $y = b$ is a horizontal line passing through the y-axis at b.

Step 2. If y is isolated on one side of the equation. Graph by plotting points.
- Choose any three values for x and then solve for the corresponding y- values.

Step 3. If the equation is of the form $Ax + By = C$, find the intercepts.
- Find the x- and y- intercepts and then a third point.

EXAMPLE 11.29

Identify the most convenient method to graph each line:

ⓐ $y = -3$ ⓑ $4x - 6y = 12$ ⓒ $x = 2$ ⓓ $y = \frac{2}{5}x - 1$

Solution

ⓐ $y = -3$

This equation has only one variable, y. Its graph is a horizontal line crossing the y-axis at -3.

ⓑ $4x - 6y = 12$

This equation is of the form $Ax + By = C$. Find the intercepts and one more point.

ⓒ $x = 2$

There is only one variable, x. The graph is a vertical line crossing the x-axis at 2.

ⓓ $y = \frac{2}{5}x - 1$

Since y is isolated on the left side of the equation, it will be easiest to graph this line by plotting three points.

TRY IT 11.56 Identify the most convenient method to graph each line:

ⓐ $3x + 2y = 12$ ⓑ $y = 4$ ⓒ $y = \frac{1}{5}x - 4$ ⓓ $x = -7$

TRY IT 11.57 Identify the most convenient method to graph each line:

ⓐ $x = 6$ ⓑ $y = -\frac{3}{4}x + 1$ ⓒ $y = -8$ ⓓ $4x - 3y = -1$

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Graph by Finding Intercepts (https://www.openstax.org/l/24findinter)

Use Intercepts to Graph (https://www.openstax.org/l/24useintercept)

State the Intercepts from a Graph (https://www.openstax.org/l/24statintercept)

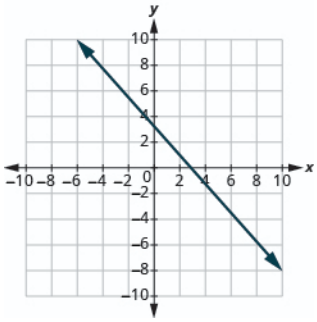
SECTION 11.3 EXERCISES

Practice Makes Perfect

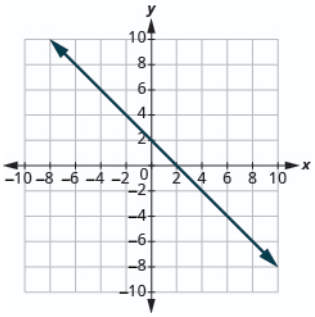
Identify the Intercepts on a Graph

In the following exercises, find the x- and y- intercepts.

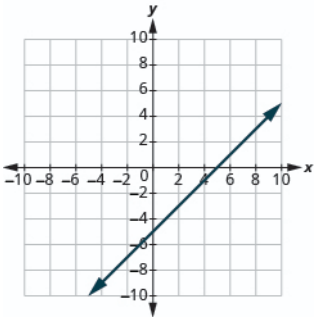
117.



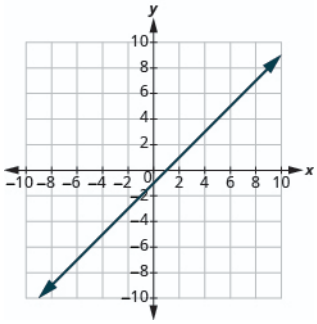
118.



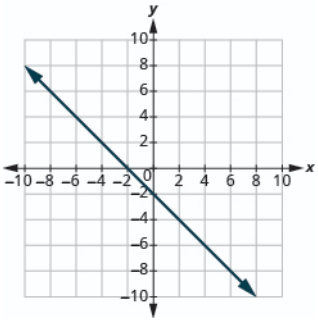
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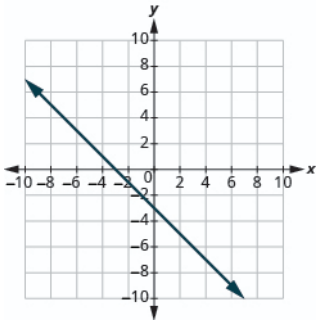
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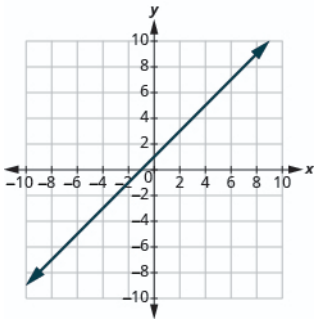
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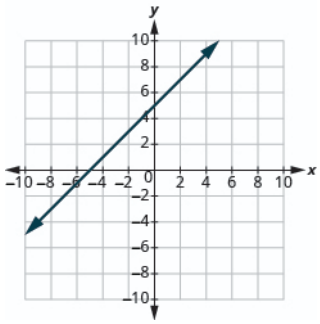
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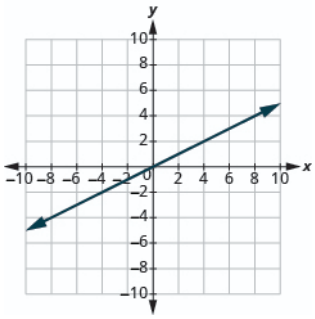
123.



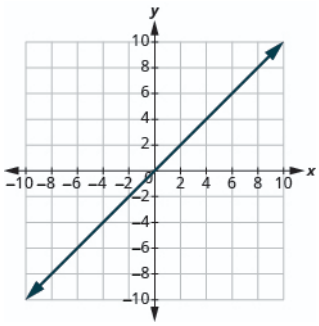
124.



125.



126.



Find the x and y Intercepts from an Equation of a Line

In the following exercises, find the intercepts.

127. $x + y = 4$

128. $x + y = 3$

129. $x + y = -2$

130. $x + y = -5$

131. $x - y = 5$

132. $x - y = 1$

133. $x - y = -3$

134. $x - y = -4$

135. $x + 2y = 8$

136. $x + 2y = 10$

137. $3x + y = 6$

138. $3x + y = 9$

139. $x - 3y = 12$

140. $x - 2y = 8$

141. $4x - y = 8$

142. $5x - y = 5$

143. $2x + 5y = 10$

144. $2x + 3y = 6$

145. $3x - 2y = 12$

146. $3x - 5y = 30$

147. $y = \frac{1}{3}x - 1$

148. $y = \frac{1}{4}x - 1$

149. $y = \frac{1}{5}x + 2$

150. $y = \frac{1}{3}x + 4$

151. $y = 3x$

152. $y = -2x$

153. $y = -4x$

154. $y = 5x$

Graph a Line Using the Intercepts

In the following exercises, graph using the intercepts.

155. $-x + 5y = 10$

156. $-x + 4y = 8$

157. $x + 2y = 4$

158. $x + 2y = 6$

159. $x + y = 2$

160. $x + y = 5$

161. $x + y = 3$

162. $x + y = -1$

163. $x - y = 1$

164. $x - y = 2$

165. $x - y = -4$

166. $x - y = -3$

167. $4x + y = 4$

168. $3x + y = 3$

169. $3x - y = -6$

170. $2x - y = -8$

171. $2x + 4y = 12$

172. $3x + 2y = 12$

173. $3x - 2y = 6$

174. $5x - 2y = 10$

175. $2x - 5y = -20$

176. $3x - 4y = -12$

177. $y = -2x$

178. $y = -4x$

179. $y = x$

180. $y = 3x$

Choose the Most Convenient Method to Graph a Line

In the following exercises, identify the most convenient method to graph each line.

181. $x = 2$

182. $y = 4$

183. $y = 5$

184. $x = -3$

185. $y = -3x + 4$

186. $y = -5x + 2$

187. $x - y = 5$

188. $x - y = 1$

189. $y = \frac{2}{3}x - 1$

190. $y = \frac{4}{5}x - 3$

191. $y = -3$

192. $y = -1$

193. $3x - 2y = -12$

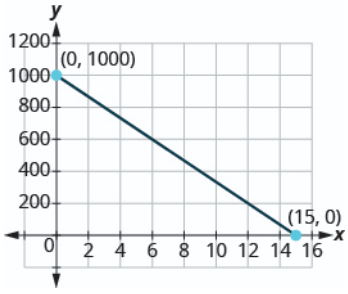
194. $2x - 5y = -10$

195. $y = -\frac{1}{4}x + 3$

196. $y = -\frac{1}{3}x + 5$

Everyday Math

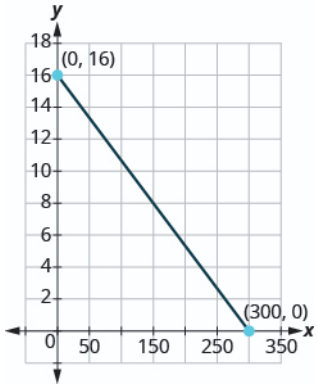
197. **Road trip** Damien is driving from Chicago to Denver, a distance of $1{,}000$ miles. The x-axis on the graph below shows the time in hours since Damien left Chicago. The y-axis represents the distance he has left to drive.



(a) Find the x- and y- intercepts

(b) Explain what the x- and y- intercepts mean for Damien.

198. **Road trip** Ozzie filled up the gas tank of his truck and went on a road trip. The x-axis on the graph shows the number of miles Ozzie drove since filling up. The y-axis represents the number of gallons of gas in the truck's gas tank.



(a) Find the x- and y- intercepts.

(b) Explain what the x- and y- intercepts mean for Ozzie.

Writing Exercises

199. How do you find the x-intercept of the graph of $3x - 2y = 6$?

200. How do you find the y-intercept of the graph of $5x - y = 10$?

201. Do you prefer to graph the equation $4x + y = -4$ by plotting points or intercepts? Why?

202. Do you prefer to graph the equation $y = \frac{2}{3}x - 2$ by plotting points or intercepts? Why?